

Qatar-1b

R-2260

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120523 · created 2026-08-16T01:12:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456070.80004 +/- 0.00021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.021
Transit depth (Rp/Rs)^2	1.8 +/- 0.56 [%]
Orbital Inclination (inc)	83.7 +/- 1.7
Airmass coefficient 1 (a1)	0.956 +/- 0.013
Airmass coefficient 2 (a2)	0.0276 +/- 0.007
Scatter in the residuals of the lightcurve fit is	1.41 %
Best Comparison Star	#2 - [265, 172]
Optimal Method	PSF photometry
Transit Duration (day)	0.063 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.134 ± 0.021 vs 0.1463 (deviation 0.59σ)
Duration fit vs published	0.063 d vs 0.0692 d (deviation 0.51σ)
Mid-time O-C	-0.4 min
Depth SNR	6.4
Residual scatter	1.41 %

Light curve

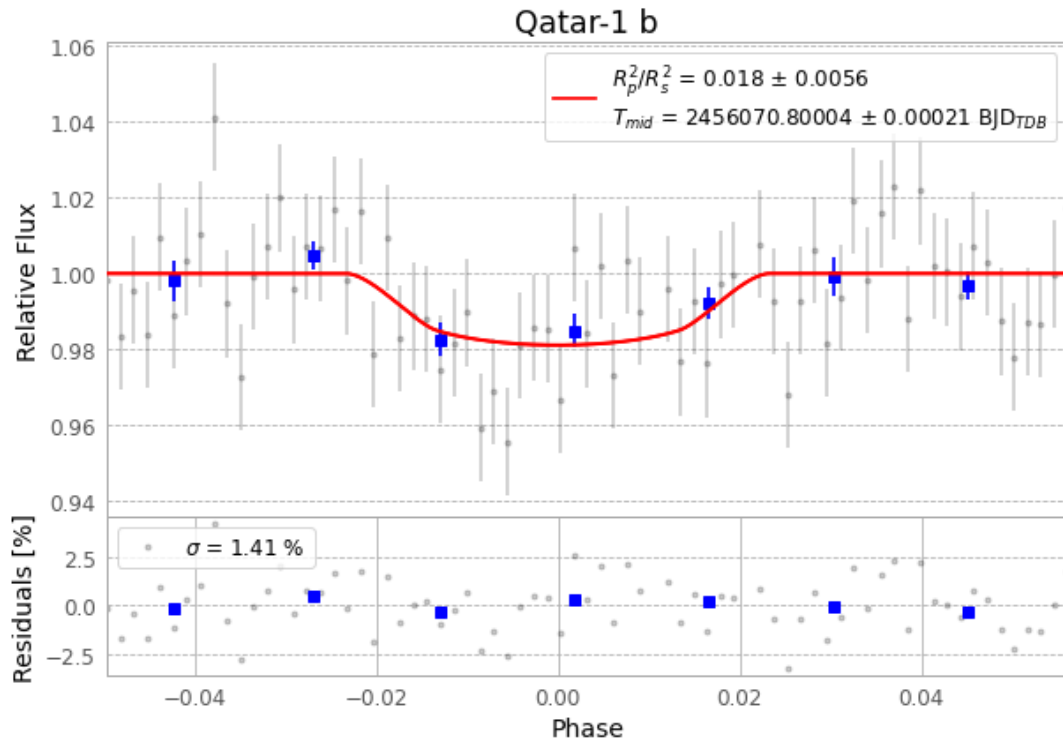


Figure 1 — FinalLightCurve_Qatar-1 b_2012-05-22.png (SHA-256 01744610c556843c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 150], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 335f44db962e8ba2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

73 FITS frames (48 MB), Qatar-1120523053012.FITS → Qatar-1120523090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120523.log (1d92a426627c...), FinalLightCurve_Qatar-1 b_2012-05-22.png (01744610c556...), FinalLightCurve_Qatar-1 b_2012-05-22.csv (6c4c0f00b15c...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:12Z.